

MTH 231 ASSESSMENT

1. The domain of the function $f(x) = \frac{1}{\sqrt{x^2 - 25}}$ is
 (A) $\{x \mid x > 5\}$ (B) $\{x \mid x < 5\}$ (C) $\{x \mid x > 25\}$ (D) $\{x \mid x < -5\} \cup \{x \mid x > 5\}$

2. $\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2} =$
 (A) 2 (B) 4 (C) 8 (D) $\sqrt{2}$

3. $\lim_{x \rightarrow 0} \frac{\sin(2x)}{x} =$
 (A) 0 (B) 1 (C) 2 (D) $\frac{1}{2}$

4. Let $f(x) = \begin{cases} 2x, & x \neq 2 \\ 0, & x = 2 \end{cases}$. Which of the following is true?
 I f is not continuous at $x = 2$ because $\lim_{x \rightarrow 2} f(x) \neq f(2)$
 II f is not continuous at $x = 2$ because $\lim_{x \rightarrow 2} f(x)$ does not exist
 III f is not continuous at $x = 2$ because 2 is not in the domain of f .
 (A) I only (B) I and II (C) II and III (D) I, II, and III

5. Let f be continuous on the closed interval $[a, b]$. Which of the following statements is/are true?
 I f attains its absolute maximum value on $[a, b]$.
 II f attains its absolute minimum value on $[a, b]$.
 III f is differentiable at each point of (a, b) .
 (A) I only (B) I and II (C) II and III (D) I, II, and III

6. Find $\frac{dy}{dx}$, where $y = (x^2 + 4)^3$.
 (A) $6x$ (B) $6x + 4$ (C) $6x(x^2 + 4)^2$ (D) $3(x^2 + 4)^2$

7. Find $\frac{dy}{dx}$, where $y = x^2 \cos x$

- (A) $x \sin x$ (B) $x \cos x$ (C) $-x^2 \sin x + 2x \cos x$ (D) $\frac{2x}{\sin x}$

8. Which of the following statements is/are true for the function $f(x) = 2x + \cos x$?

I f is strictly increasing.

II f is strictly decreasing.

III f does not have critical numbers.

- (A) I only (B) I and II (C) I and III (D) II and III

9. An equation of the tangent line to the graph of $y = x^2 + 1$ at the point where $x = 1$ is

- (A) $Y = 2x$ (B) $y = x + 1$ (C) $x + y = 1$ (D) $y = 2(x + 1)$

10.

x	0	1	2	3	4
$f(x)$	-2	2	0	-2	2
$f'(x)$	9	0	-1	0	3

The table above gives values of f and f' at selected points in the interval $[0, 4]$.

Answer TRUE or FALSE.

_____ f has a relative maximum at $x = 1$.

_____ f has a relative minimum at $x = 3$.

_____ The absolute minimum value of f on $[0, 4]$ is -2 .

11. At which points on the graph of f are f' and f'' both negative?

12. Find an equation of the normal line to the graph of $x^3 + y^3 = 3xy$ at the point $\left(\frac{3}{2}, \frac{3}{2}\right)$.

In exercises 13 – 19, $f(x) = x^3 - 12x$.

13. Find the critical points of f .
14. Find the intervals where f is increasing.
15. Find the interval where f is decreasing.
16. Find the relative extrema of f .
17. Find the coordinates of inflection point on the graph of f .
18. Determine the interval where the graph of f is concave upward.
19. Determine the interval where the graph of f is concave downward.

For exercises 20 – 22.

A particle moves along the x -axis so that its displacement (distance) from the origin at time t is given by $x(t) = t^2 - 3t + 2$, where x is in feet and t in seconds.

20. Find the velocity?
21. For what positive value(s) of t is the particle motionless?
22. Use the differential of the function $f(x) = \sqrt[3]{x}$ to approximate $\sqrt[3]{8.02}$.
23. Verify that $y = \cos(2x)$ is a solution of the equation $\frac{d^2y}{dx^2} + 4\cos(2x) = 0$.